Regularity and processing complexity in recursive numeral systems

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Experimental and computational work have demonstrated that languages culturally evolve in response to simplicity and informativeness (Kirby, Tamariz, Cornish, & Smith, 2015; Carlsson, Dubhashi, & Regier, 2024). This is in line with recent work in computational typology, which shows that natural linguistic systems lie close to the *Pareto frontier*, the theoretical optimal bound beyond which a system cannot improve on one measure without sacrificing the other (Kemp, Xu, & Regier, 2018; Gibson et al., 2019). Interestingly, however, systems of recursive numerals (e.g., English numerals like *one*, *two*, ..., *twenty-nine*, *thirty*, ...) are all maximally informative as they can precisely express any integer (Xu, Liu, & Regier, 2020). As such, Denić and Szymanik (2024) (D&S) argued that such systems instead optimise the trade-off between two aspects of complexity: lexicon size and average morphosyntactic complexity (which they operationalise as average number of morphemes per numeral). While they have shown that natural recursive numeral systems are (near-)Pareto-optimal under these two pressures, other systems which are Pareto-optimal under their measures exhibit unnatural properties, such as having alternating bases (e.g., $5 * \mathbf{4}$ for 20, $3 * \mathbf{7}$ for 21, $\mathbf{25} - 3$ for 22, $\mathbf{25} - 2$ for 23, $6 * \mathbf{4}$ for 24, **25** for 25). Yang and Regier (2025) (Y&R) attempted to solve this issue with ad-hoc constraints that restrict possible systems to be more natural, e.g. by enforcing the same base to be used for consecutive numerals, but their constraints are not motivated by a general principle.

In this study, we argue instead that the relevant trade-off governing recursive numeral systems is between *regularity* and *processing complexity*. Our formalisation of these measures is broadly couched in the Minimum Description Length framework (Li, Vitányi, et al., 2008), as used in work on the evolution of compositionality (Brighton, 2003). To measure regularity, we programmatically infer the minimal partial deterministic finite automaton (DFA) that could generate all and only the numerals (in the range 1-99, for tractability) in a system using an established algorithm (Mihov & Schulz, 2019). Intuitively, a regular system like a natural one will feature reuse across numerals (e.g., a base-10 system like Mandarin will reuse the structure $X * 10 + Y$), resulting in a smaller minimal

DFA. Adapting Brighton (2003), then, our measure of *ir*regularity is the encoding length (in bits) of the inferred DFA. As shown in Brighton (2003), this measure of regularity trades off with *processing complexity*, i.e., the average number of bits required to encode a unique parsing path through the DFA. Informally, this measures the number of cognitive decisions (the state transitions) one has to make to parse or generate a numeral morpheme-by-morpheme.

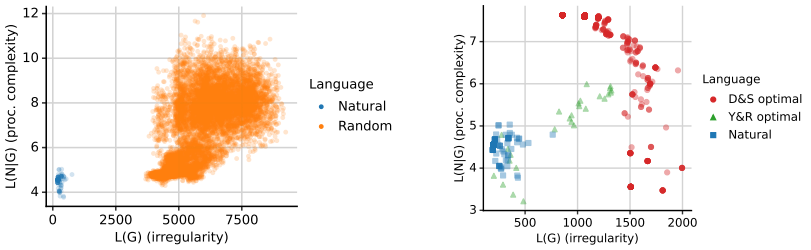

Figure 1.: Irregularity and processing complexity of natural systems (blue squares). **Left:** vs. random possible systems (orange). **Right:** vs. optimal possible languages in D&S (red circles) and Y&R (green triangles).

With these measures, we show that natural systems exhibit significantly higher regularity and lower processing complexity than other theoretically possible systems. As Figure 1 (left) shows, natural systems are significantly more efficient under our measures than other theoretically possible systems that use digits and bases from natural languages.[1] Crucially, as Figure 1 (right) shows, natural systems are significantly more efficient than the vast majority of the optimal systems from the two previous works, and the artificial systems that occupy the same area in the space as the natural ones are precisely the ones that are human-like due to Y&R's ad-hoc constraints. Taken together, these results suggest that our measures, unlike the previous ones, capture the key characteristics of natural recursive numeral systems that distinguish them from unnatural ones, and that Y&R's ad-hoc constraints follow naturally from the measures.

While language evolution has explored cross-linguistic properties of recursive numerals (Hurford, 1975) and their evolution (Hurford, 1987, 1999), our approach

[1] An anonymous reviewer questions whether there is a trade-off, since natural systems are very clustered, and distinct from artificial systems (e.g., in Figure 1, left). We do not yet explicitly identify a Pareto frontier here (this would require exploring the full space, which we leave for future work), however an analogous trade-off has been shown generally in Brighton (2003). It is also worth noting that continuity is not necessary for a trade-off (e.g., one could imagine a space that has a gap in the middle but nevertheless has a Pareto frontier). In this case, the gap likely reflects the fact that natural systems are difficult to sample; our algorithm could, in theory, sample a typical, natural base-10 numeral system, but it is so rare that it is not in our random sample. We also note that the full space may still be discontinuous because of arithmetical constraints on top of the trade-off (i.e., morphemes combinations must evaluate to all integers).

is a first step to building more explanatorily powerful evolutionary models. For example, while previous simulations assume Y&R-like constraints to evolve regular, human-like systems (Hurford, 1987), our measures could be used to show how such systems could evolve through agents' interactions without explicit constraints (as in Silvi, Thomas, Carlsson, Dubhashi, & Johansson, 2025). Moreover, our approach, which incorporates the regularity of forms, could potentially be applied to any linguistic domain that makes use of regular form-meaning mappings (i.e., compositional systems, such as person systems; Zaslavsky, Maldonado, & Culbertson, 2021), an area that has been under-explored in existing work, which often treats signals as having no internal structure (cf. Osmelak, Xu, Hahn, & McCurdy, 2026). We wish to extend our work in both directions.

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